



BITS Pilani
Pilani Campus



SURVEYING (CE F213)

(Compass Surveying)

Compass surveying

- The meaning of compass survey, types of compass
- Concept of bearing
- In compass survey, the **direction of the survey line** is measured by the use of a magnetic compass while the lengths are by chaining or taping.
- Where the area to be surveyed is comparatively large, the compass survey is preferred, whereas if the area is small in extent and a high degree of accuracy is desired, then chain survey is adopted.
- However, where the compass survey is used, care must be taken to make sure that magnetic disturbances are not present.
- The two major primary types of survey compasses are: the **prismatic compass** and **surveyors compass**.

Difference between direction and angle

Direction: is a measured angle with respect to a fixed reference.

(Surveyor's compass, prismatic compass,...)

Angle: measured angle without any fixed reference.

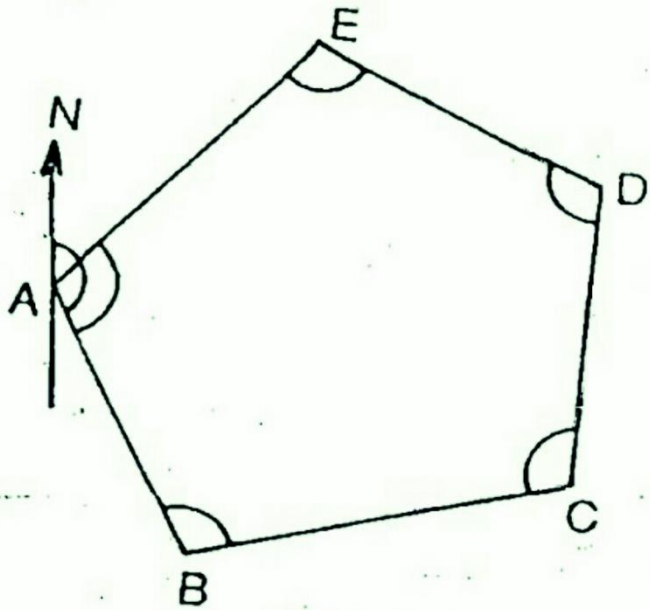
(Sextant, Theodolite,...)

Method adopted

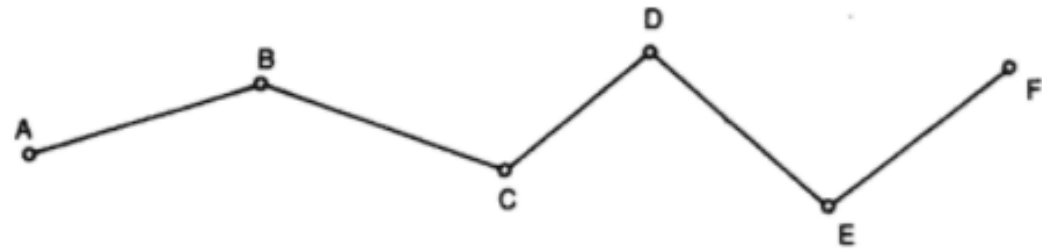
The compass survey is also carried out making **traverse**, like the chain surveying.

Closed traverse: When a series of connected lines forms a closed circuit, i.e. when the finishing point coincides with the starting point of a survey, it is called as a 'closed traverse'

Open traverse: When a sequence of connected lines extends along a general direction and does not return to the starting point, it is known as 'open traverse' or (unclosed traverse).



Closed Traverse



Open Traverse

Measurement of angles

It can be done in 2-ways.

- (i) In relation to each other(angle).
- (ii) In relation to a fixed reference. [The fixed reference is called **meridian** and the angle measured is called **bearing**]

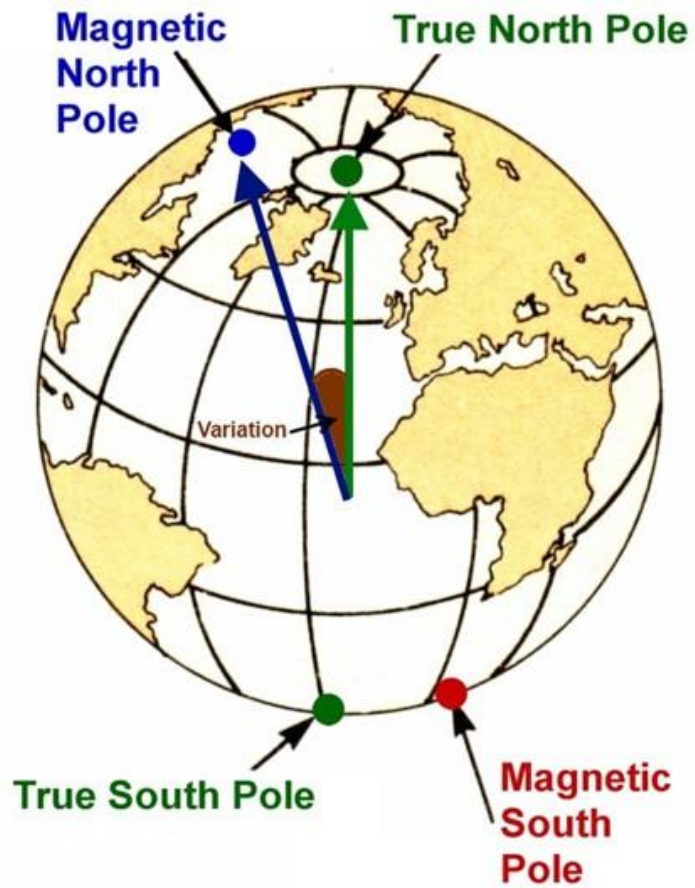
Meridian and bearing

Meridian (It is a fixed direction with respect to which the bearings of survey lines are expressed)

1. True meridian (Geographical meridian)
2. Magnetic meridian
3. Grid meridian
4. Arbitrary meridian

True meridian: The true meridian at a given point on earth surface is a line(part of the imaginary circle) passing through the true north pole, true south pole and that point.

What is longitude and what is latitude at a point?



Magnetic meridian: The magnetic meridian at any place is the direction obtained by observing the position of a freely supported magnetized needle when it comes to rest uninfluenced by local attracting forces.

Grid meridian: The grid meridian is the reference meridian for a country on a national survey map.

Arbitrary meridian: It is any convenient direction from the survey station to any well-defined permanent object.

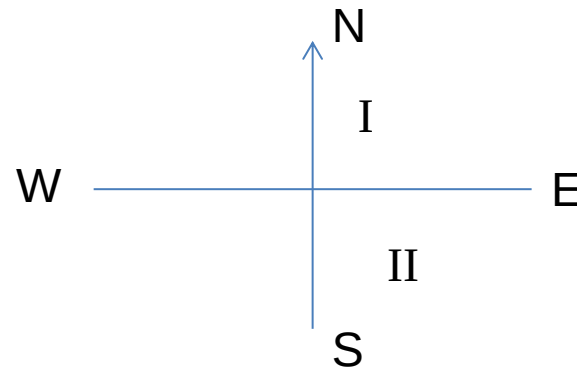
Bearing: (It is the horizontal angle between the reference meridian and survey line measured clockwise or anticlockwise direction)

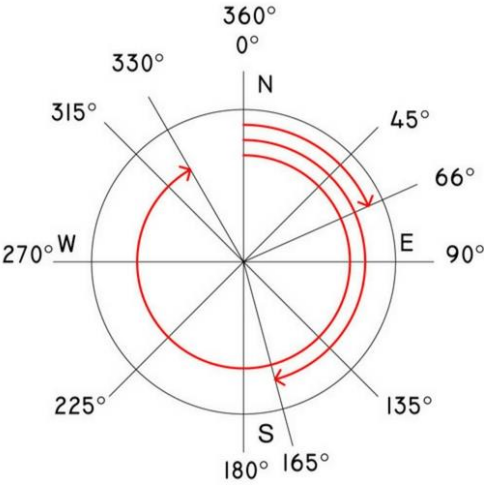
1. True bearing
2. Magnetic bearing
3. Grid bearing
4. Arbitrary bearing

Presentation of Bearing

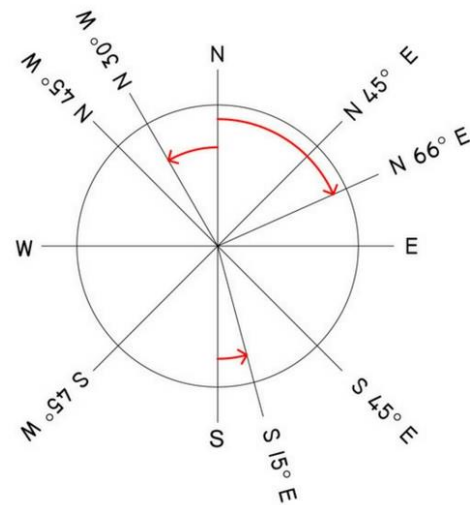
1. Whole Circle Bearing(WCB),
2. Quadrant Bearing
3. Reduced Bearing

WCB	Quadrant	RB	QB
0 TO 90	I	$RB = WCB$	N-E
90 TO 180	II	$RB = 180 - WCB$	S-E
180 TO 270	III	$RB = WCB - 180^\circ$	S-W
270 TO 360	IV	$RB = 360^\circ - WCB$	N-W

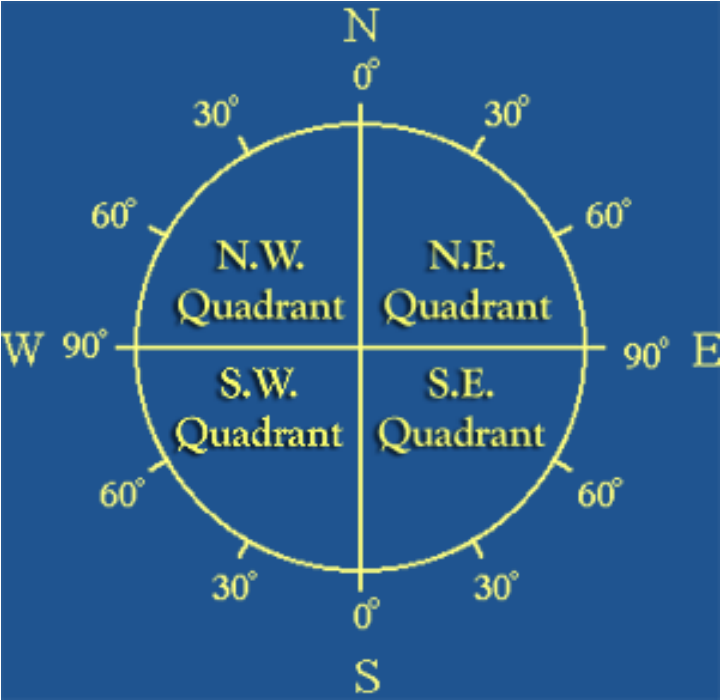




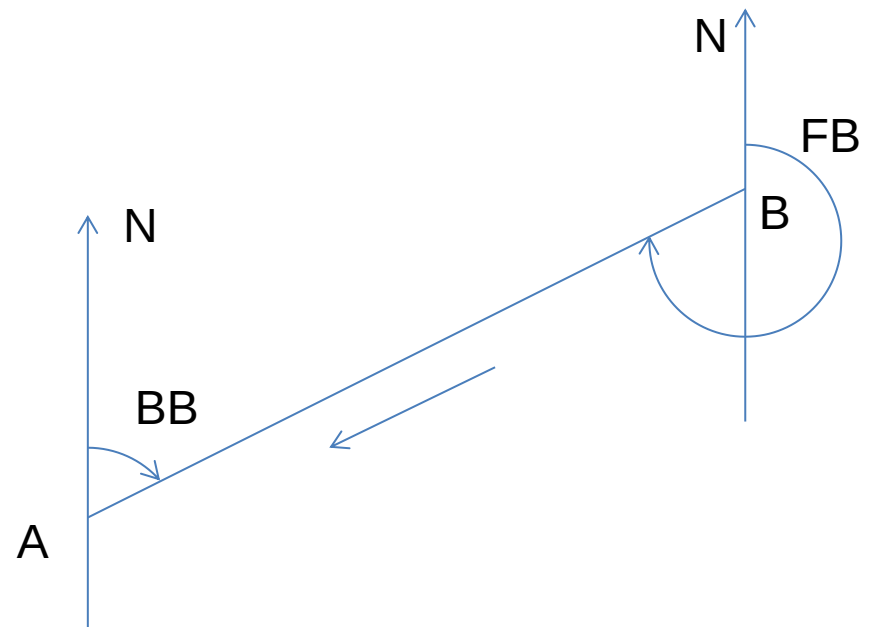
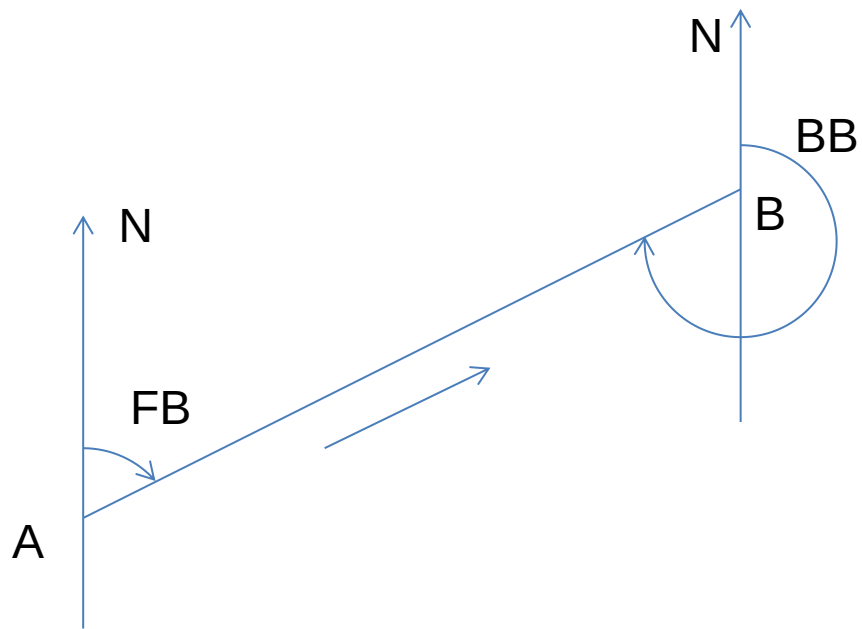
Whole Circle Bearing



Quadrantal Bearing



Fore Bearing(FB) and Back Bearing(BB)



Back Bearing = Fore Bearing $\pm 180^0$

+ve when the fore bearing is less than 180^0

-ve when the fore bearing is more than 180^0

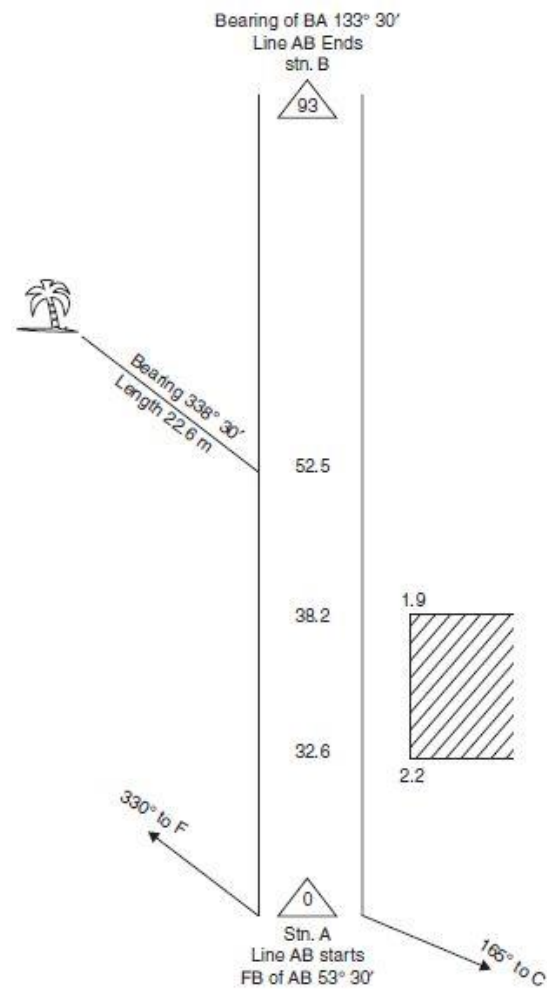


Fig. 13.12

Units we take for measuring angle

(a) Sexagesimal System (Relating to 60 and its base)

1 circumference = 360°

1 degree = 60 minutes = $60'$

1 minute = 60 seconds = $60''$

(b) Hour System

(c) Centesimal System

Types of Compass



PRISMATIC COMPASS



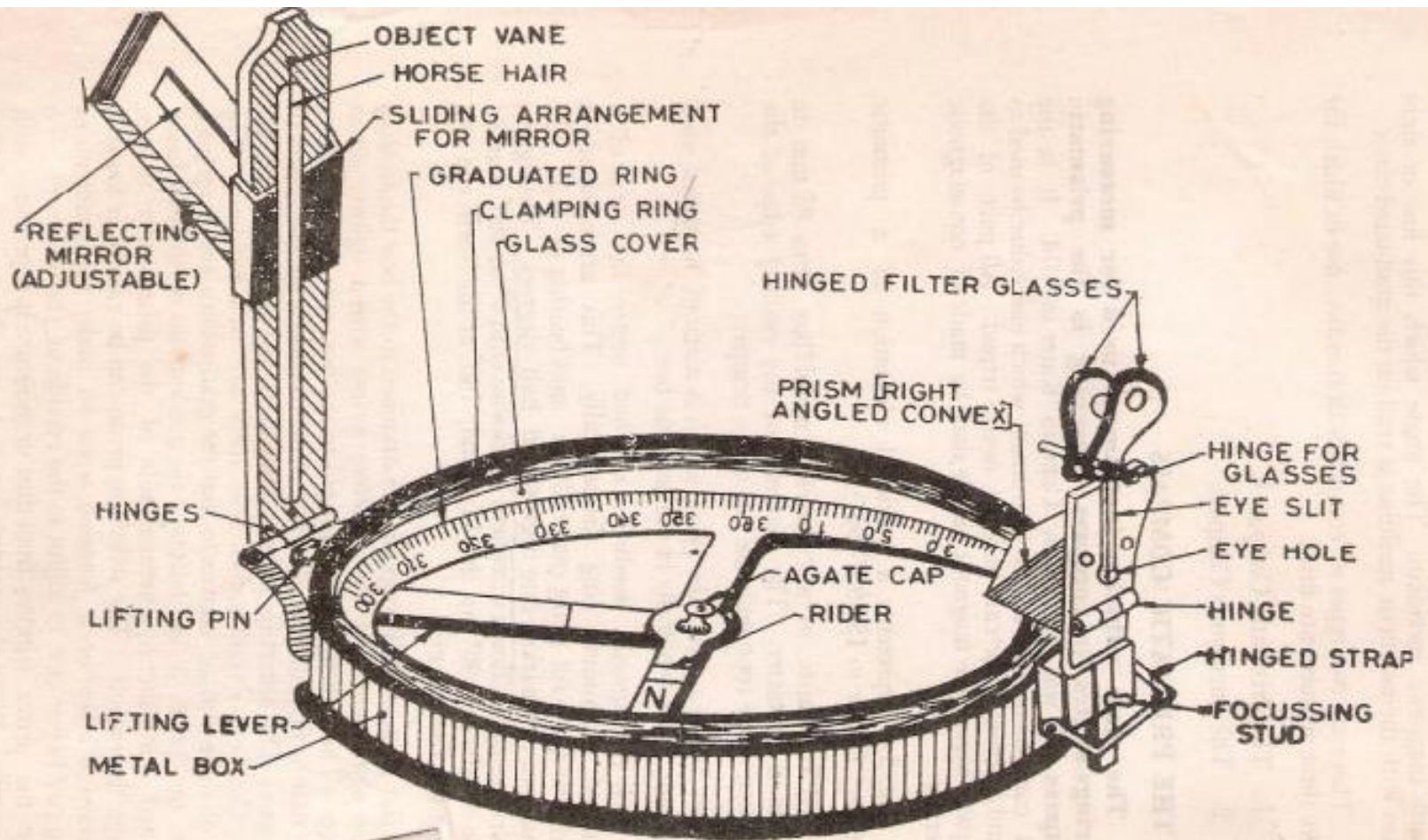
SURVEYOR'S COMPASS

Prismatic Compass

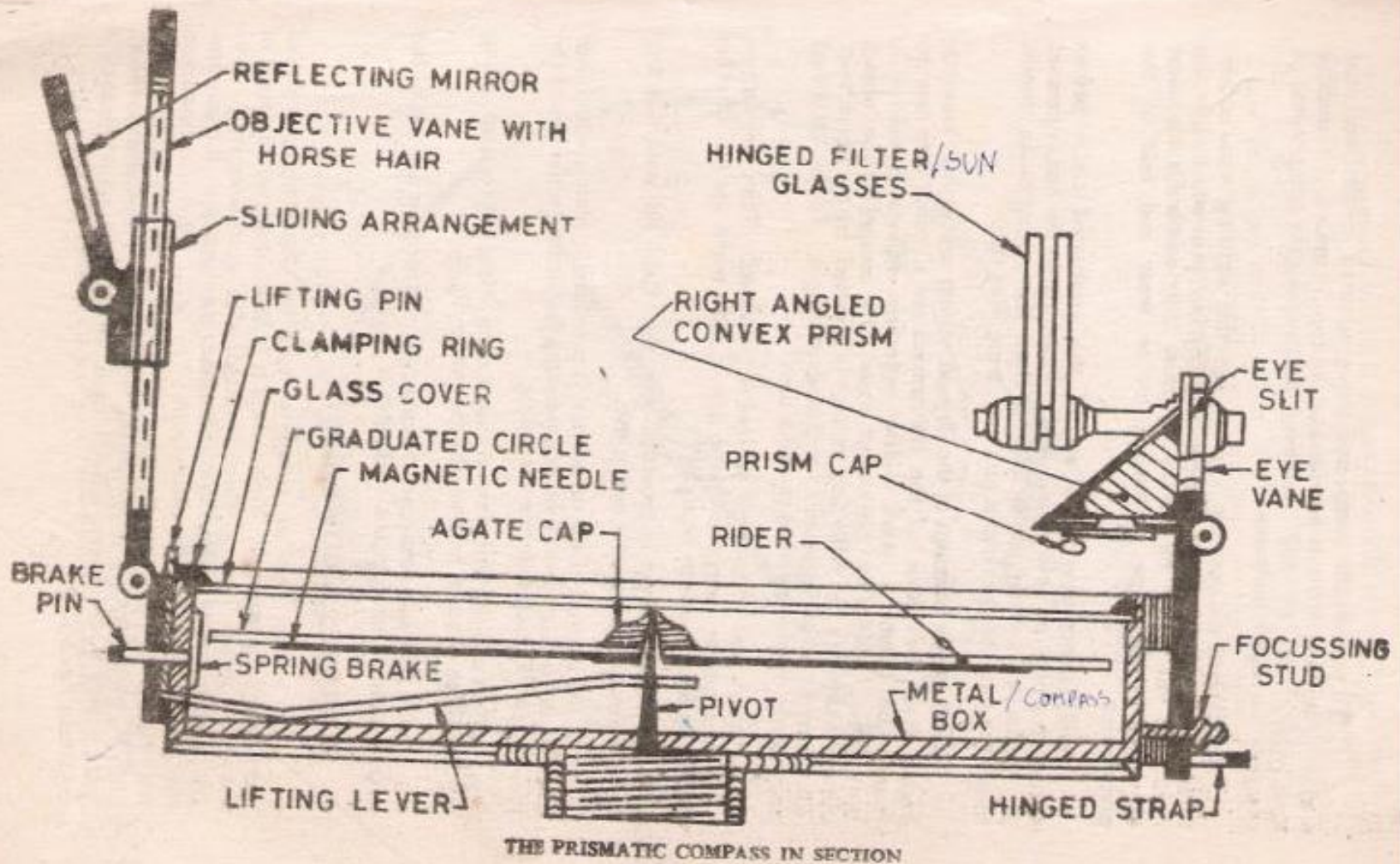
- Graduation circle is fixed to broad type needle. Hence, it will not rotate with the line of sight.
- There is a prism at viewing end.
- Sighting and reading can be done simultaneously.
- The magnetic needle do not act as an index.
- The graduations are in whole circle bearing.
- Graduations are marked inverted since its reflection is read through prism.
- The reading is taken through a prism.
- Tripod may or may not be used. It can be held on a stretched hand also.

Surveyors Compass

- Graduation circle is fixed to the box. Hence, it rotates with the line of sight.
- At viewing end there is no prism. There is only a slit.
- Sighting and viewing cannot be done simultaneously.
- Magnetic needle acts as index while reading.
- The graduations are in quadrantal system.
- Graduations are marked directly. They are not inverted.
- The reading is taken by directly viewing from top glass.
- Tripod is essential for using it.



THE PRISMATIC COMPASS SHOWING ITS VARIOUS PARTS



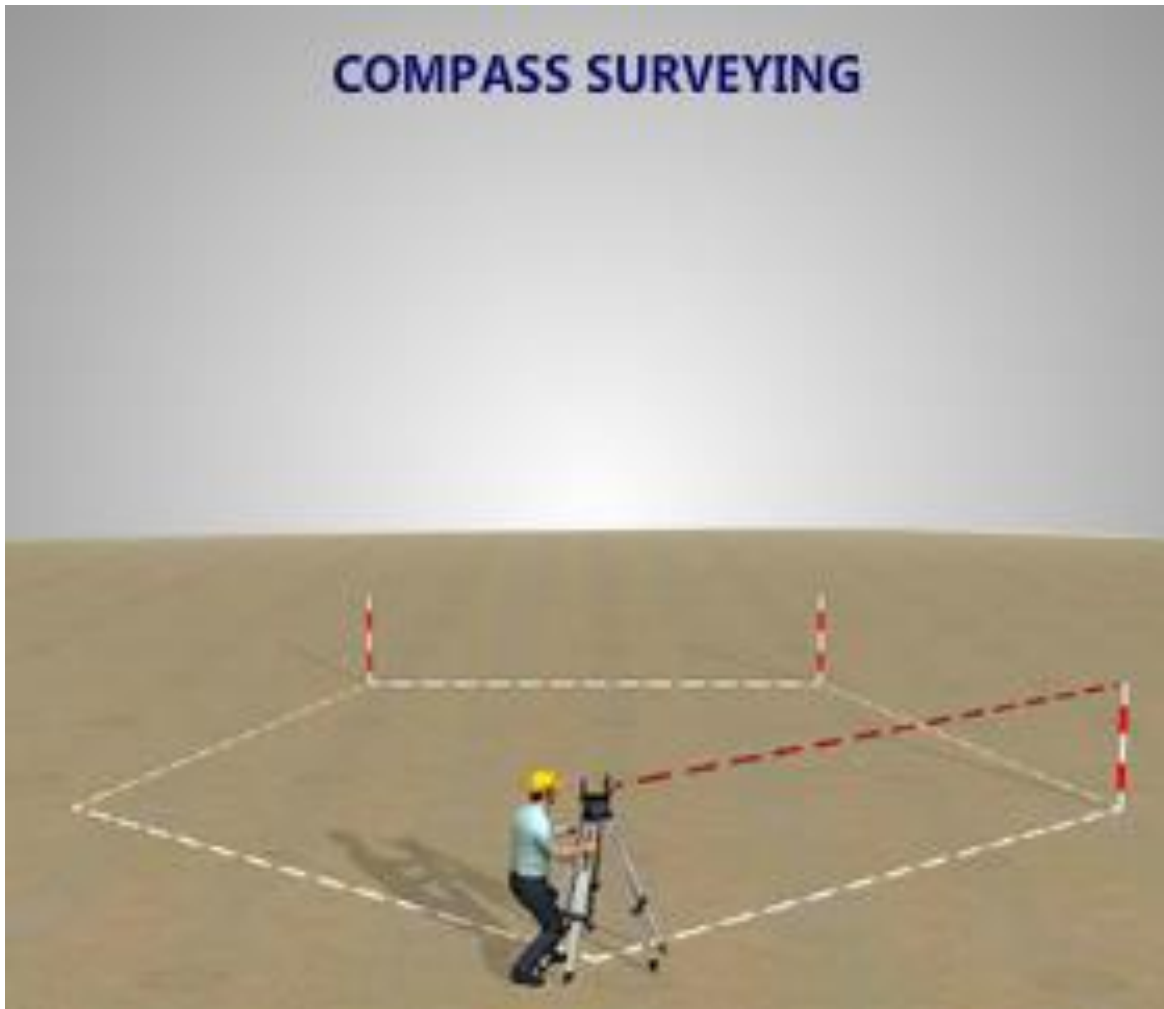
Temporary adjustment of prismatic compass



The following procedure should be adopted after fixing the prismatic compass on the tripod for measuring the bearing of a line.

- **Centering** : Centering is the operation in which compass is kept exactly over the station from where the bearing is to be determined. The centering is checked by dropping a small pebble from the underside of the compass. If the pebble falls on the top of the peg then the centering is correct, if not then the centering is corrected by adjusting the legs of the tripod.
- **Leveling** : Leveling of the compass is done with the aim to freely swing the graduated circular ring of the prismatic compass. The ball and socket arrangement on the tripod will help to achieve a proper level of the compass. This can be checked by rolling round pencil on glass cover.
- **Focusing** : The prism is moved up or down in its slide till the graduations on the aluminum ring are seen clear, sharp and perfect focus. The position of the prism will depend upon the vision of the observer.

COMPASS SURVEYING

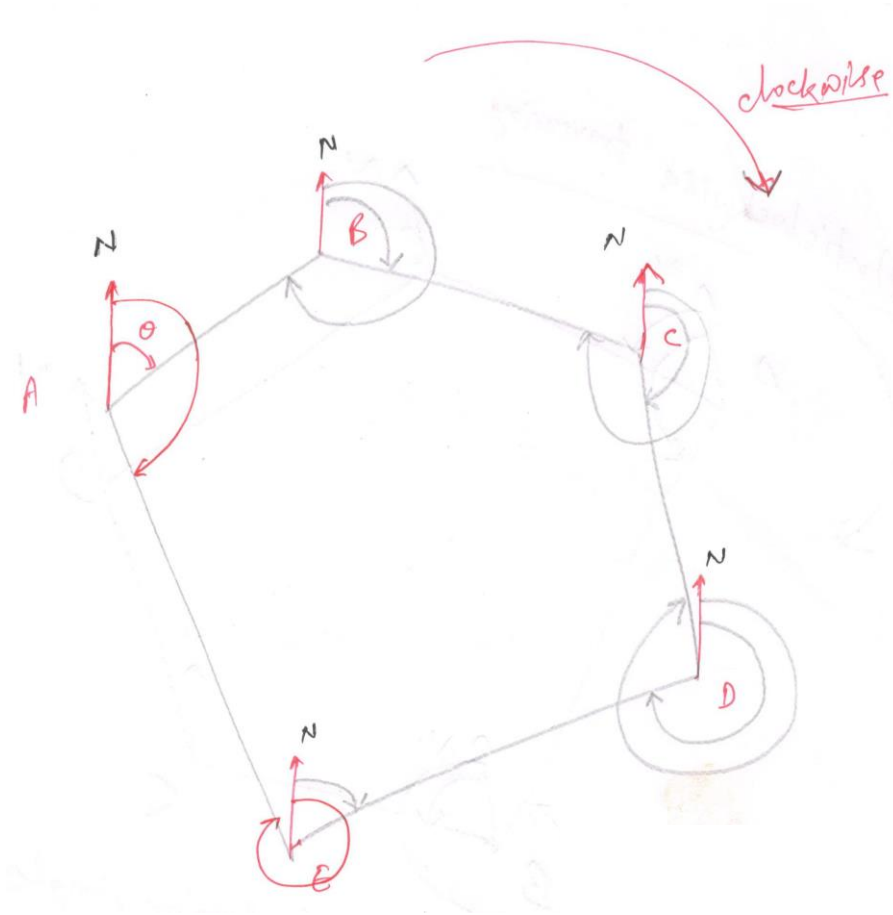


Magnetic declination

Included angle: Can be external can be internal.

1. Clockwise traverse-
2. Anticlockwise traverse-

- Clockwise Traverse-



Included angle at A = FB of forward line - BB of previous line
= FB of line AB - BB of line EA (If this is -ve, we have to add
360°)

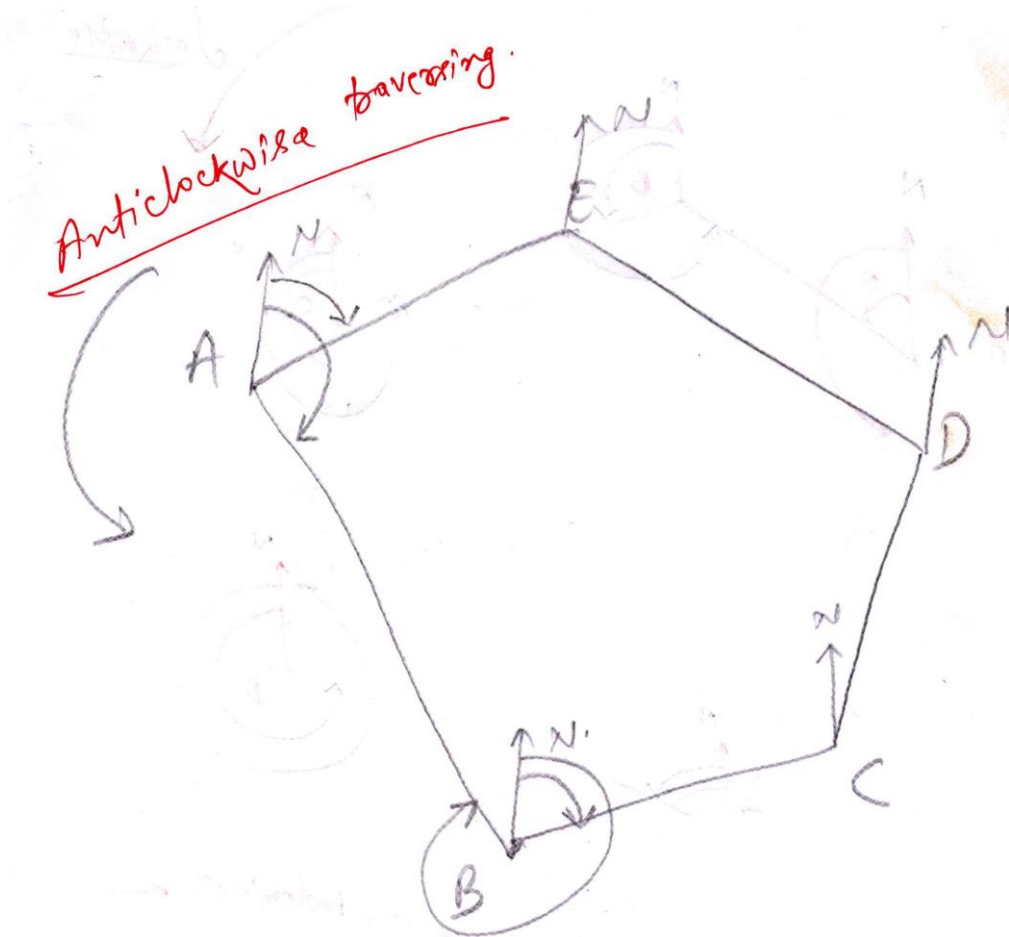
(In this we shall get **exterior** included angles)

OR

Included angle at A = BB of previous line - FB of forward line
= BB of line EA - FB of line AB (If this is -ve, we have to add
360°)

(In this we shall get **Interior** included angles)

- Anticlockwise traverse-



Included angle at A = FB of forward line - BB of previous line
= FB of line AB - BB of line EA (If this is –ve, we have to add
360°)

(In this we shall get **interior** included angles)

OR

Included angle at A = BB of previous line - FB of forward line
= BB of line EA - FB of line AB (If this is –ve, we have to add
360°)

(In this we shall get **exterior** included angles)

Errors in Compass surveying

Local Attraction

- A magnetic needle points towards the magnetic north pole if suspended or pivoted. Local attraction is the influence that prevents magnetic needle pointing to magnetic north pole.
- Unavoidable substance that affect are
 - Magnetic ore
 - Underground iron pipes
 - High voltage transmission line
 - Electric pole etc.
- Influence caused by avoidable magnetic substance doesn't come under local attraction such as instrument, watch wrist, key etc.

Q1. A closed traverse ABCDA with the following detail exists. Find the included angles and state whether these are exterior or interior.

Line	FB(WCB)
AB	40°
BC	50°
CD	118°
DA	250°

Solution

Q2. The bearings observed in a traverse survey conducted by prismatic compass is presented in table below. The local attraction is suspected during the survey.

Line	FB(WCB)	BB(WCB)
AB	124° 30'	304° 30'
BC	68° 15'	246° 00'
CD	310° 30'	135° 15'
DA	200° 15'	17° 45'

Find the stations affected by local attraction and correct the bearings.

Solution(method-I) (Applying corrections to bearings)

Back Bearing = Fore Bearing $\pm 180^\circ$

+ve when the fore bearing is less than 180°

-ve when the fore bearing is more than 180°

Station A and Station B are free from local attraction.

Line BC

The FB of line BC is correct, which $68^\circ 15'$,

So the BB of BC should be $248^\circ 15'$.

So the error is $2^{\circ} 15'$ (-ve error) , at station C

Line CD

FB of CD is $310^{\circ} 30'$ which incorrect,

Correct FB of CD is $312^{\circ} 45'$

Correct BB of CD is $312^{\circ} 45' - 180^{\circ} = 132^{\circ} 45'$

But the observed BB of CD is $135^{\circ} 15'$, so error is $2^{\circ} 30'$ (+ve error), it is at station D

Line DA

FB of DA is $200^{\circ} 15'$ which is having error,

Correct FB of DA is $200^{\circ} 15' - 2^{\circ} 30' = 197^{\circ} 45'$,

So the BB of DA is $197^{\circ} 45' - 180^{\circ} = 17^{\circ} 45'$ (same as observed)

Solution Method –II (Included angle method)

To calculate all included angles: All **interior** angles are

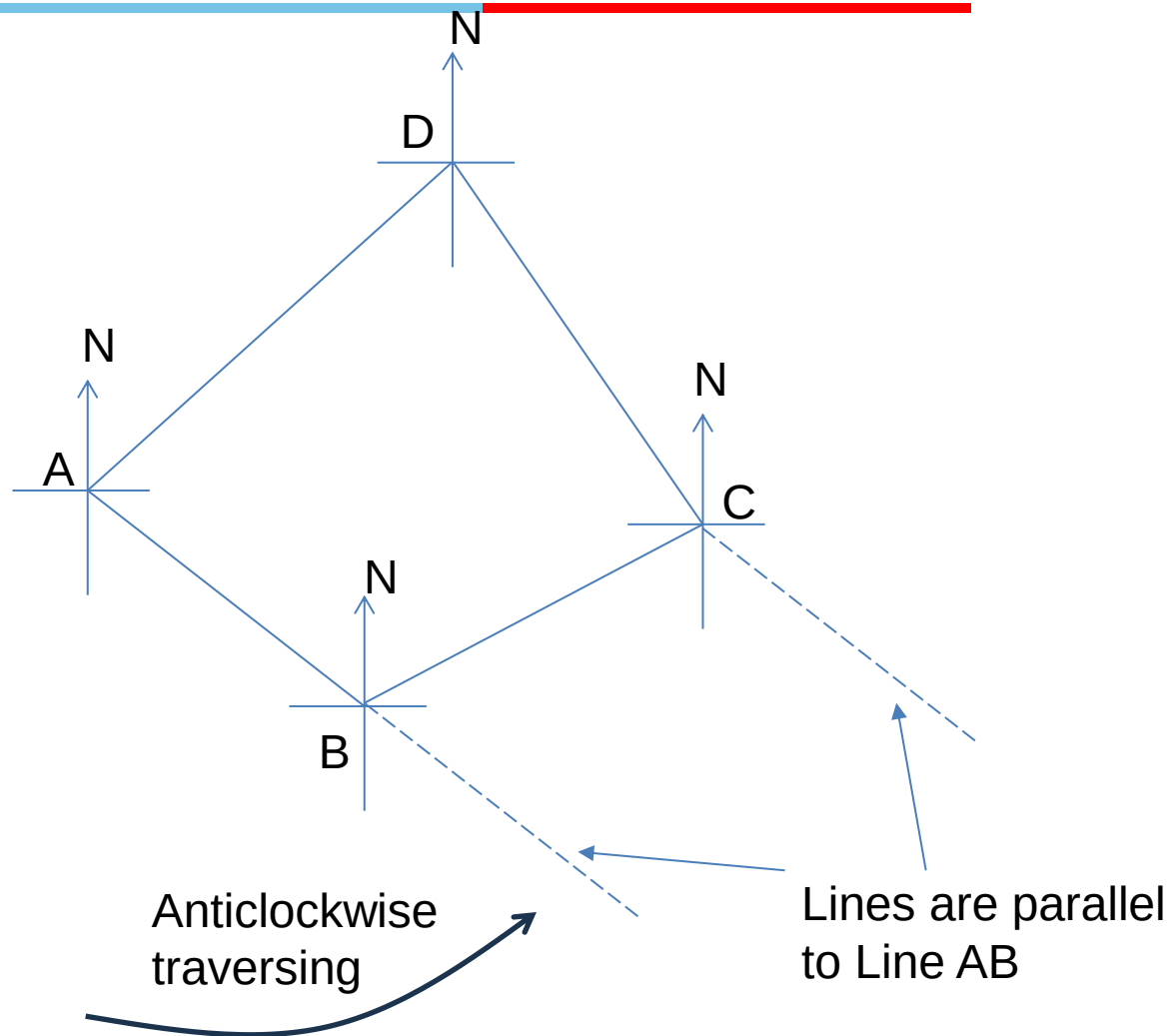
$$\text{Angle A} = 106^{\circ} 45'$$

$$\text{Angle B} = 123^{\circ} 45'$$

$$\text{Angle C} = 64^{\circ} 30'$$

$$\text{Angle D} = 65^{\circ} 00'$$

To check sum of all interior included angle = Angle A + Angle B + Angle C + Angle D = 360°



Line BC

FB of line BC = $68^{\circ} 15'$ (it is correct)

$$\begin{aligned}\text{BB of line BC} &= \text{FB of line AB (previous line)} + \text{included angle at B} \\ &= 124^{\circ} 30' + 123^{\circ} 45' \\ &= 248^{\circ} 15'\end{aligned}$$

Line CD

$$\begin{aligned}\text{BB of line CD} &= \text{FB of line BC (previous line)} + \text{included angle at C} \\ &= 68^{\circ} 15' + 64^{\circ} 30' = 132^{\circ} 45'\end{aligned}$$

$$\begin{aligned}\text{FB of line CD} &= \text{BB of line CD} + 180^\circ \\ &= 132^\circ + 180^\circ \\ &= 312^\circ 45'\end{aligned}$$

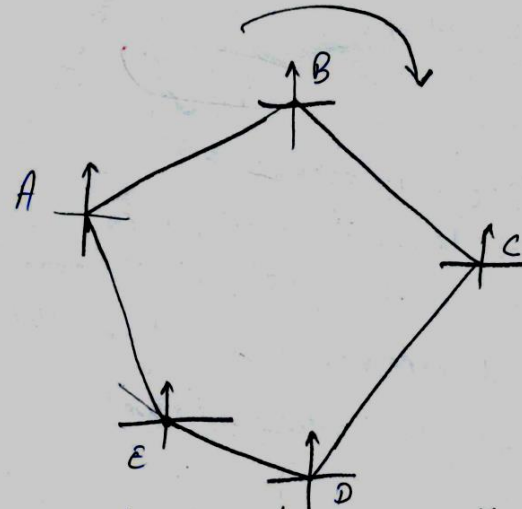
Line DA

$$\begin{aligned}\text{BB of line DA} &= 17^\circ 45' \text{ (it is correct)} \\ \text{FB of Line DA} &= 17^\circ 45' + 180^\circ \\ &= 197^\circ 45'\end{aligned}$$

Q3: The fore bearing and back bearing of the lines in a traverse conducted with the prismatic compass is mentioned in the table. All the bearings are whole circle bearings. The local attraction and observation error is suspected. Using the included angle method find the correct bearings of the lines.

Line	FB	BB
AB	35° 35'	213° 43'
BC	135° 18'	316° 10'
CD	200° 34'	19° 34'
DE	250° 36'	72° 39'
EA	320° 53'	141° 18'

No-3



It is a clockwise traverse. In this the included angle will be exterior angles using the following expressions.

Exterior Included angles.

$$\begin{aligned} \angle A &= \text{FB of } AB - \text{BB of } EA \\ &= 35^\circ 35' - 141^\circ 18' = -105^\circ 43' \\ &\approx -105^\circ 43' + 360^\circ \\ &= 254^\circ 17' \end{aligned}$$

$$\begin{aligned} \angle B &= \text{FB of } BC - \text{BB of } AB \\ &= 135^\circ 18' - 213^\circ 43' = -78^\circ 25' \\ &\approx -78^\circ 25' + 360^\circ \\ &= 281^\circ 35' \end{aligned}$$

$$\begin{aligned} \angle C &= \text{FB of } CD - \text{BB of } BC = 200^\circ 34' - 316^\circ 10' \\ &= -115^\circ 36' \approx 244^\circ 24' \end{aligned}$$

$$\begin{aligned} \angle D &= \text{FB of } DE - \text{BB of } CD = 250^\circ 36' - 19^\circ 34' \\ &= 231^\circ 2' \end{aligned}$$

int. $(128^\circ 58')$

innovate

achieve

lead

2.5) $\angle E = \text{FB of } EA - \text{BB of } DE = 320^\circ 53' - 72^\circ 39'$
 $\text{int. } 111^\circ 14' = 248^\circ 14'$

To check the sum of included angles.

$$\angle A + \angle B + \angle C + \angle D + \angle E = 1259^\circ 32'$$

Actual sum should be $(2n+4)90^\circ = 1260^\circ$

Error is -ve and it is $28'$. So the error is to be distributed to each angle, so $5'36''$ to be added to each angle.

So the corrected exterior included angles are

$$\angle A = 254^\circ 22' 36''$$

$$\angle B = 281^\circ 40' 36''$$

$$\angle C = 244^\circ 29' 36''$$

$$\angle D = 231^\circ 7' 36''$$

$$\angle E = 248^\circ 19' 36''$$

To check which line is having correct bearings for having minimum error.

Line	FB	BB	FB ~ BB	Error
AB	$35^\circ 35'$	$213^\circ 43'$	$178^\circ 8'$	$-1^\circ 52'$
BC	$135^\circ 18'$	$316^\circ 10'$	$180^\circ 52'$	$+52'$
CD	$200^\circ 34'$	$19^\circ 34'$	181°	$+1^\circ$
DE	$250^\circ 36'$	$72^\circ 39'$	$177^\circ 57'$	$-2^\circ 3'$
EA	$320^\circ 53'$	$141^\circ 18'$	$179^\circ 35'$	$-25'$

Let's choose line EA, as it is having minimum error in ~~deviation~~ difference between F.B. and B.B. 25' will be distributed in both bearings.

So F.B. of EA = $320^{\circ}53' + \frac{25'}{2} = 321^{\circ}5'30''$ (9)
 B.B. of EA = $141^{\circ}18' - \frac{25'}{2} = 141^{\circ}5'30''$ (2)

Line AB

F.B. of AB = B.B. of EA + A
 $= 141^{\circ}5'30'' + 254^{\circ}22'36'' = 395^{\circ}28'6''$
 $\approx 35^{\circ}28'6''$
 B.B. of AB = $35^{\circ}28'6'' + 180^{\circ} = 215^{\circ}28'6''$ (1)

Line BC

F.B. of BC = B.B. of AB + B
 $= 215^{\circ}28'6'' + 281^{\circ}40'36'' = 497^{\circ}8'42''$
 $= 137^{\circ}8'42''$
 B.B. of BC = $137^{\circ}8'42'' + 180^{\circ} = 317^{\circ}8'42''$ (1)

Line CD

F.B. of CD = B.B. of line BC + C = $317^{\circ}8'42'' + 244^{\circ}27'18''$
 $= 561^{\circ}38'18''$
 $\approx 201^{\circ}38'18''$
 B.B. of CD = $201^{\circ}38'18'' - 180^{\circ} = 21^{\circ}38'18''$ (1)

Line DE

F.B. of line DE = B.B. of line CD + D
 $= 21^{\circ}38'18'' + 231^{\circ}7'36''$
 $= 252^{\circ}45'54''$
 B.B. of line DE = $252^{\circ}45'54'' - 180^{\circ} = 72^{\circ}45'54''$

Line EA

F.B. of line EA = B.B. of line DE + E
 $= 72^{\circ}45'54'' + 248^{\circ}19'36''$
 $= 321^{\circ}5'30''$ checked (1)

This Question can be solved by finding the internal included angles also. All the answers will match.

